MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET - A) (SOLUTIONS)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM:** 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. Partial marks are allocated step-wise strictly according to CBSE board evaluation norms as shown in brackets.
3. Every proof highlights stated assumptions, substitutions, and concluding justification.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ $(p \neq 0)$ is: [1]

(A) 1 (B) $\frac{1}{p}$ (C) -1 (D) $-\frac{1}{p}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) -1** [0.5 Mark]

Using the common difference definition:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$d = \frac{1-p}{p} - \frac{1}{p} = \frac{1-p-1}{p} = \frac{-p}{p} = \mathbf{-1}$.

Q2. If the nth term of an AP is $7 - 4n$, then its common difference is: [1]

(A) 4 (B) -4 (C) 3 (D) 11

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) -4** [0.5 Mark]

Evaluating consecutive terms a_1 and a_2:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$a_1 = 7 - 4(1) = 3$ and $a_2 = 7 - 4(2) = -1 \implies d = -1 - 3 = \mathbf{-4}$.

Q3. If the common difference of an AP is 5, then what is $a_{18} - a_{13}$? [1]

(A) 10 (B) 20 (C) 25 (D) 30

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 25** [0.5 Mark]

Using the nth term formula:

Formula / Theorem Used: $a_m - a_n = (m-n)d$

Justification: [0.5 Mark]

$a_{18} - a_{13} = (a + 17d) - (a + 12d) = 5d = 5(5) = \mathbf{25}$.

Q4. **Assertion (A):** Three numbers a, b, c are in AP if and only if $2b = a + c$.
Reason (R): The sum of the first n odd natural numbers is n^2. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: (B) Both (A) and (R) are true; (R) is not correct explanation. [0.5 Mark]

Evaluating each statement independently:

Formula / Theorem Used: $b - a = c - b \iff 2b = a + c$ vs. $S_n^{\text{odd}} = \frac{n}{2}[2 + 2(n-1)] = n^2$

Justification: [0.5 Mark]

Assertion (A) is true by constant difference; Reason (R) is also true as an independent identity, so (R) does not explain (A). Hence option **(B)**.

Q5. **Assertion (A):** The sum of the first 15 terms of the AP $21, 18, 15, 12, \ldots$ is zero ($S_{15} = 0$).
Reason (R): The sum of the first n terms of an AP is $S_n = \frac{n}{2}[2a + (n-1)d]$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: (A) Both (A) and (R) are true, and (R) explains (A). [0.5 Mark]

Applying the sum formula from Reason (R) with $a = 21, d = -3, n = 15$:

Formula / Theorem Used: $S_{15} = \frac{15}{2}[2(21) + (15-1)(-3)] = \frac{15}{2}(42 - 42) = 0$

Justification: [0.5 Mark]

Both statements are true, and substituting the AP parameters into (R) directly proves Assertion (A).

Q6. Find the 9th term from the end (towards the first term) of the AP $5, 9, 13, \ldots, 185$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying l and d [1 Mark]

$a = 5$, $d = 9 - 5 = 4$, last term $l = 185$, $n = 9$.

Formula / Theorem Used: n^{th} term from end $= l - (n-1)d$

Step 2: Computing 9th Term from End [1 Mark]

$$9^{\text{th}} \text{ term from end} = 185 - (9-1)(4) = 185 - 8 \times 4 = 185 - 32 = \boxed{\mathbf{153}}$$

Q7. How many two-digit numbers are divisible by 3? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Forming the AP [1 Mark]

Two-digit numbers divisible by 3: $12, 15, 18, \ldots, 99$ ($a = 12$, $d = 3$, $a_n = 99$).

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 2: Solving for n [1 Mark]

$$12 + (n-1)(3) = 99 \implies 3(n-1) = 87 \implies n - 1 = 29 \implies \boxed{\mathbf{n = 30}}$$

Q8. If 7 times the 7th term of an AP is equal to 11 times its 11th term, find its 18th term. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $7a_7 = 11a_{11}$. Let first term $= a$, common difference $= d$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Expanding $7a_7 = 11a_{11}$ [1 Mark]

$7(a + 6d) = 11(a + 10d) \implies 7a + 42d = 11a + 110d$.

Step 2: Simplifying to Obtain a_{18} [1 Mark]

$11a - 7a + 110d - 42d = 0 \implies 4a + 68d = 0 \implies a + 17d = 0 \implies \boxed{\mathbf{a_{18} = 0}}$.

Q9. Find the sum of the first 30 terms of the AP $-30, -24, -18, \ldots$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying Parameters [1 Mark]

$a = -30$, $d = -24 - (-30) = 6$, $n = 30$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 2: Calculating S_{30} [1 Mark]

$$S_{30} = \frac{30}{2}[2(-30) + (30-1)(6)] = 15(-60 + 174) = 15 \times 114 = \boxed{\mathbf{1{,}710}}$$

Q10. If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$, find its 15th term (a_{15}). [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $S_n = 3n^2 + 5n$.

Formula / Theorem Used: $a_{15} = S_{15} - S_{14}$

Step 1: Evaluating S_{15} and S_{14} [1 Mark]

$S_{15} = 3(15)^2 + 5(15) = 3(225) + 75 = 675 + 75 = 750$.

$S_{14} = 3(14)^2 + 5(14) = 3(196) + 70 = 588 + 70 = 658$.

Step 2: Finding a_{15} [1 Mark]

$$a_{15} = 750 - 658 = \boxed{\mathbf{92}}$$

Q11. The ratio of the sums of the first n terms of two APs is $(7n+1) : (4n+27)$. Find the ratio of their m^{th} terms. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Representing Ratio of Sums of First n Terms [1.5 Marks]

Let the first terms be a_1, a_2 and common differences be d_1, d_2.

Formula / Theorem Used: $\dfrac{S_n}{S'_n} = \dfrac{2a_1+(n-1)d_1}{2a_2+(n-1)d_2} = \dfrac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2}$

$$\frac{\frac{n}{2}[2a_1+(n-1)d_1]}{\frac{n}{2}[2a_2+(n-1)d_2]} = \frac{7n+1}{4n+27} \implies \frac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27} \quad \text{— (i)}$$

Step 2: Matching Coefficients for the Ratio of mth Terms [2 Marks]

The ratio of mth terms is $\frac{T_m}{T'_m} = \frac{a_1+(m-1)d_1}{a_2+(m-1)d_2}$. Equating coefficient of d_1, d_2 in (i):

$$\frac{n-1}{2} = m-1 \implies n-1 = 2m-2 \implies n = 2m-1$$

Step 3: Substituting $n = 2m-1$ Into (i) [1.5 Marks]

$$\frac{T_m}{T'_m} = \frac{7(2m-1)+1}{4(2m-1)+27} = \frac{14m-7+1}{8m-4+27} = \boxed{\frac{\mathbf{14m-6}}{\mathbf{8m+23}}}$$

Q12. How many terms of the AP $45, 39, 33, \ldots$ must be taken so that their sum is 180? Explain the double answer. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Forming Quadratic Equation in n [1.5 Marks]

Here $a = 45$, $d = 39-45 = -6$, $S_n = 180$.

Formula / Theorem Used: $S_n = \dfrac{n}{2}[2a+(n-1)d]$

$$\frac{n}{2}[2(45)+(n-1)(-6)] = 180 \implies \frac{n}{2}[96-6n] = 180 \implies n(48-3n) = 180$$

Step 2: Solving by Middle-Term Splitting [2 Marks]

Dividing by 3: $n(16-n) = 60 \implies n^2-16n+60 = 0$.

Splitting middle term $(-6-10=-16, (-6)(-10)=60)$:

$$n^2-6n-10n+60 = 0 \implies n(n-6)-10(n-6) = 0 \implies (n-6)(n-10) = 0 \implies n = 6 \text{ or } 10$$

Step 3: Explaining the Double Answer [1.5 Marks]

Since $d = -6 < 0$, terms from a_7 to a_{10} cancel out identically: $a_7 = 9$, $a_8 = 3$, $a_9 = -3$, $a_{10} = -9$.

Because $a_7 + a_8 + a_9 + a_{10} = 9+3-3-9 = 0$, both 6 terms and 10 terms give sum 180.

$$\boxed{\mathbf{n = 6 \text{ or } n = 10}}$$

Q13. The houses of a row are numbered consecutively from 1 to 49. Show that there is a value of X such that the sum of numbers of the houses preceding the house numbered X is equal to the sum of the numbers of the houses following it. Find this value of X. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Translating House Sums Preceding and Following X [1.5 Marks]

Houses preceding X: $1, 2, \ldots, (X-1)$ with sum S_{X-1}. Houses following X: $(X+1), \ldots, 49$ with sum $S_{49} - S_X$.

Formula / Theorem Used: Condition: $S_{X-1} = S_{49} - S_X$, where $S_k = \dfrac{k(k+1)}{2}$

Step 2: Algebraic Reduction [2 Marks]

$$\frac{(X-1)X}{2} = \frac{49 \times 50}{2} - \frac{X(X+1)}{2} \implies X^2 - X = 2{,}450 - X^2 - X \implies 2X^2 = 2{,}450$$

Step 3: Solving and Validating Bounds ($1 \le X \le 49$) [1.5 Marks]

$$X^2 = \frac{2{,}450}{2} = 1{,}225 \implies X = \sqrt{1{,}225} = \boxed{\mathbf{35}} \quad \text{(rejecting } X = -35\text{)}$$

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